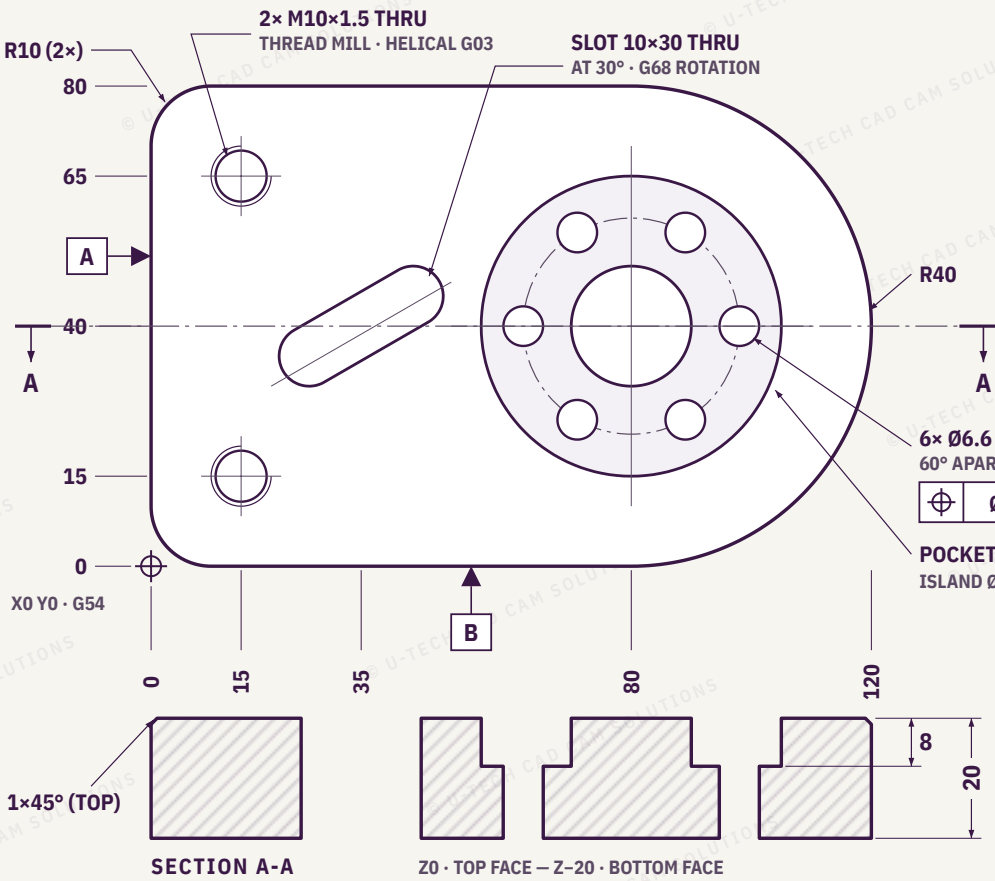
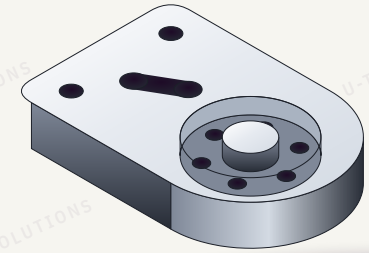
VIEW 01 PLAN VIEW · SECTION A-A — Ordinate dims from G54 datum · mm



VIEW 02 ISOMETRIC
Rendered · not to scale



SETUP · G54

X0 Y0 — bottom-left corner · Z0 top face
Stock 125 × 85 × 23, vice on parallels

ASSIGNMENT 3 · ADVANCED

1. **Tangent-arc contour R40/R10 with G02/G03**
2. **Island pocketing: G41 inside, G42 round island**
3. **PCD pattern by G16 polar or M98 subprogram**
4. **G68 rotation for the 30° slot**
5. **Thread milling M10 by helical G03 (Z per rev)**

GENERAL NOTES

01 Material: AL 6061-T6, stock 125 × 85 × 23.
02 All dimensions in mm, from datum X0 Y0.
03 Untoleranced dimensions: ISO 2768-mK.
04 Break edges 0.2×45°; 1×45° on top edge only.
05 Contour: G41; R40 by G02 with R, R10 by G03.
06 Pocket: Ø12 rough +0.3, Ø8 finish, island G42.
07 PCD: G16 X18 Y0...300, or M98 P100 L6 + G68.
08 Slot: G68 X35 Y40 R30, mill straight, G69.

STUDENT _____ ROLL NO. _____ DATE _____ MARKS / 50 · P20 S8 W15 R7

FEATURE TABLE

G54 · mm

ID	X	Y	FEATURE	Z
H1	15	15	M10×1.5	THRU
H2	15	65	M10×1.5	THRU
H3–8	80	40	6×Ø6.6 PCD36	THRU
P1	80	40	PKT Ø50	–8
I1	80	40	ISLAND Ø20	0
S1	35	40	SLOT 10×30 @30°	THRU
C1	80	40	R40 END, R10×2	–20.5

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TITLE

CAM PLATE

ASSIGNMENT 3 · ADVANCED

DRG. NO. UTCCS-VMC-A3-003	SHEET 1 / 1
MATERIAL AL 6061-T6	STOCK 125 × 85 × 23
SCALE NTS · UNIT mm	PROJECTION 1st angle
GENERAL TOL. ISO 2768-mK	SURFACE Ra 3.2 (√)

LIMITS OF SIZE

FEATURE	FIT	UPPER	LOWER
ISLAND Ø20	h8	0.000	–0.033
PKT Ø50	H9	+0.062	0.000
PCD Ø36	—	+0.030	–0.030

REQUIRED TOOLING

VMC · 3-AXIS

T01	Face top	Ø50 face mill
T02	Contour · pocket rough	Ø12 end mill
T03	Island finish · slot	Ø8 end mill
T04	Spot · PCD holes G81	Ø6.6 drill
T05	M10 tap drill G83	Ø8.5 drill
T06	Thread mill, helical G03	M10×1.5 TM
T07	Top edge chamfer	Ø10 chamfer 45°

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